



Department of Information Technology
Academic Year 2024–2025 (Even Semester)

Degree, Semester & Branch : IV Semester, B.Tech, IT
Course Code / Title : CS3491/ Artificial Intelligence and Machine Learning
Name of the Faculty member : Mr. G. Mareeswari

Innovative Practice Description

- **Unit/ Topic:** Unit V - Activation functions
- **Course Outcome:** CO5
- **Activity Chosen:** Flipped Classroom
- **Justification:**

The flipped classroom approach for the topic *Activation Function* in Neural Networks promotes active and personalized learning by enabling students to explore foundational concepts—such as types of activation functions (e.g., sigmoid, tanh, ReLU, etc), their characteristics, mathematical formulations, and roles in learning—prior to the class through ppt shared in canvas earlier. This independent exploration allows students to understand the significance of activation functions in introducing non-linearity to neural networks and enabling complex learning patterns. Overall, the flipped classroom cultivates a deeper, and interactive session for understanding of activation function. It also provides Peer-to-peer discussions lead to richer insights and improved conceptual clarity.

- **Time Allotted for the Activity:** 20Minutes
- **Details of the Implementation:**
 - Relevant learning resources such as PPT was uploaded to Canvas two days in advance to allow students early access.
 - Students were instructed to review the content independently and note down key points based on their understanding.
 - During the in-class session, learners were grouped into teams of four and given 10 minutes for collaborative discussion on the topic.

- Following that, a representative from each team presented their collective insights to the whole class, as depicted in Figures 1 and 2.
- CO–PO/ PSO mapping:**

CO	PO1	PO3	PO9	PO10	PSO1	PSO3
CO5	2	1	3	2	1	1

(1– Low 2 –Moderate 3 – High)

- PO/ PSO mapped:**

Innovative practice	PO1	PO3	PO9	PO10	PSO1	PSO3
CO5	2	1	3	2	1	1
Justification for correlation	Needs basic engineering knowledge for understand the concept of Activation function.	The students can able to find solution for the complex ML problems using the Activation function.	Students involved in this activity as an individual and also as a team.	Communicate/ share the ideas with other students	Activation function is used in many data science projects to solve complex computational problem	The students can use this concept of Activation function in software project development.

- Images/ Screenshot of the practice:**



Figure 1

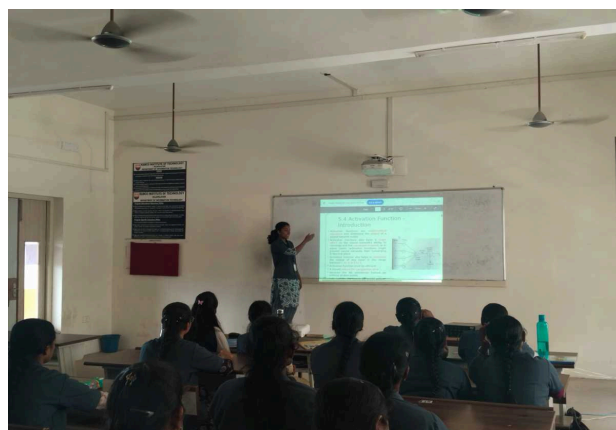


Figure 2

Figure 1 and 2 shows the Glimpses of presentation done by Sibiah. R during the flipped class room Activity.

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- Students reported a better understanding of activation functions after peer discussions.
- Students expressed that it helped them become more aware of their own learning styles.

- ❖ ***Benefit of the practice:***

- It encourages active student participation and critical thinking.
- It helps students to enhance conceptual clarity through peer interaction
- It allows for more personalized support from the teacher.
- It strengthens collaborative skills and promotes knowledge sharing.
- It helps students independent learning habits and better time management.

- ❖ ***Challenges faced in implementation:***

- Some students found it hard to understand technical content without direct instruction.
- Not all students contributed equally in group tasks, affecting collaboration.

References:

- ❖ <https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20independently%20at%20home.>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [youtube.com/watch?v=BCIxikOq73Q](https://www.youtube.com/watch?v=BCIxikOq73Q)
- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
- ❖ https://www.ritrjpm.ac.in/images/B.Tech-IT/2023-2024 / IP/ GM_CS3491 _Flipped_Classroom.pdf

Signature of Faculty Member

HOD